Katabasis | Markets Research and System Design

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https://katabasis.rip/

Research ownership and integrated scope

Katabasis is my independent markets research project. I define the financial questions, select data and methods, design how the components work together, and evaluate the findings. AI generates the software implementation from my specifications.

The system connects market-data collection and source research with hypothesis testing, options replay, portfolio analysis and evidence review. Each study links a financial question to explicit assumptions, comparisons and decision criteria.

Selected research and design work

01 A framework for financial hypotheses

Defined research standards around economic mechanisms, counterparty incentives, market constraints and capacity. Specified the data requirements and tests that would distinguish a proposed effect from an unsupported explanation.

02 Comparing portfolio products by their role

Compared eight income and cash-management ETFs by portfolio role, liquidity, fees and historical drawdown. Incorporated differences in inception history into risk scenarios with explicit coverage assumptions.

03 Evaluating the trade-offs in portfolio controls

Evaluated 26 configurations of volatility targeting, drawdown controls and rebalancing bands under three transaction-cost assumptions. Identified a band that met the study's historical tracking and turnover criteria while examining growth and drawdown trade-offs.

04 Using a placebo to investigate data integrity

Used placebo tests and source checks to investigate an apparent return pattern. Traced the effect to incompatible price adjustments and required consistent inputs for the corrected analysis.

Explore the public project

System design and research workflows
https://katabasis.rip/system/

Documented case: data mismatch in a return calculation
https://katabasis.rip/work/data-mismatch/